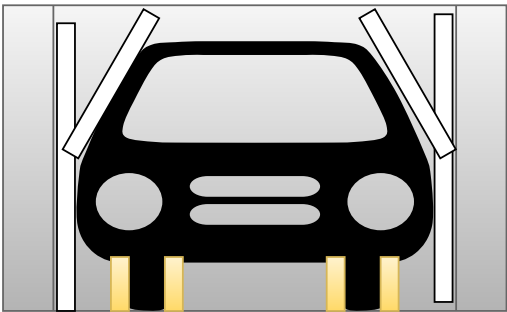
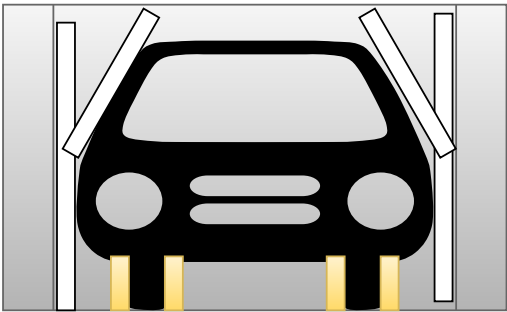
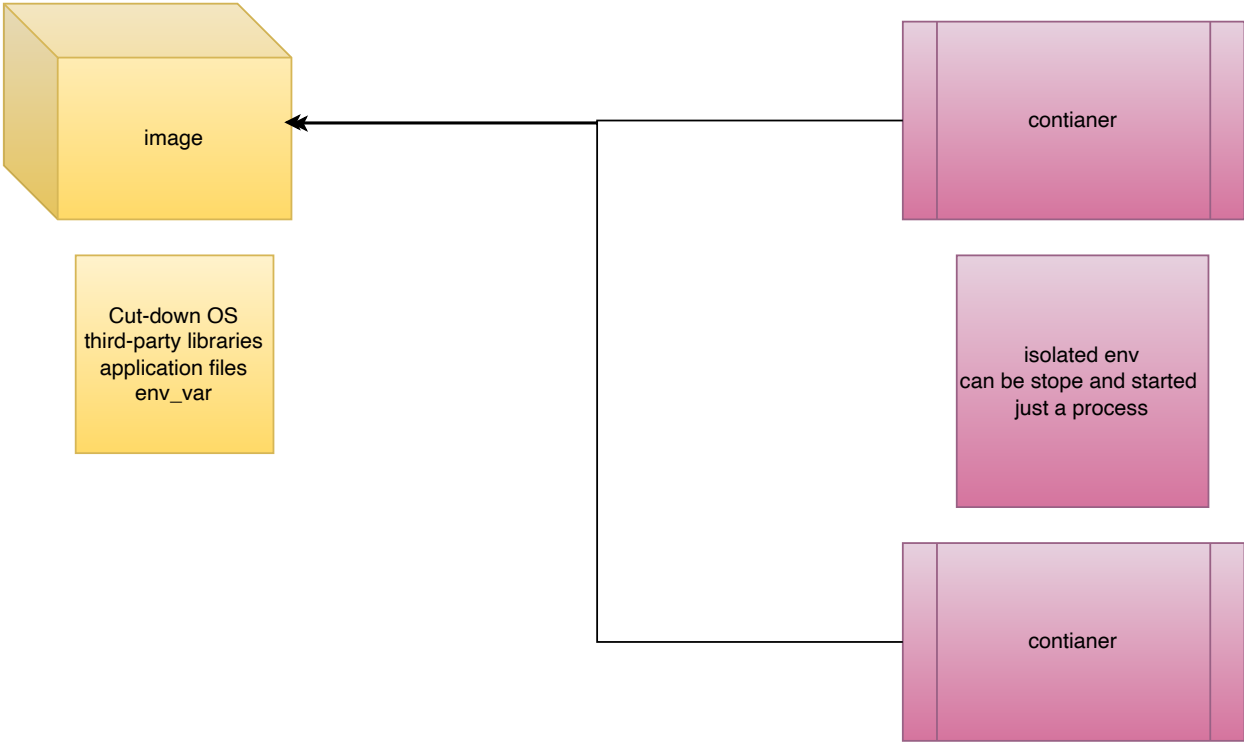


Docker images



shipping agreement

contianer must be in a black color

the wheels must be tied to the ground

the wheels must be at 30 degree form the door

the door must be locked

~~~~~

~~~~~

??

?

docker

why docker ?

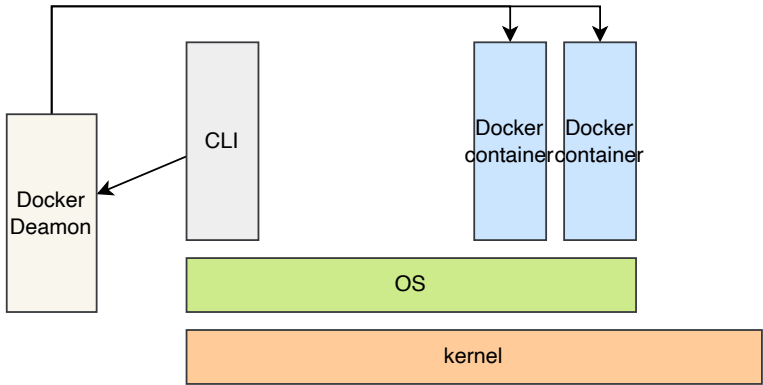
- consistency : works on every machine
- isolation: application run in isolated env
- efficiency : lightweight os
- scalability : ease to scale

reasons:

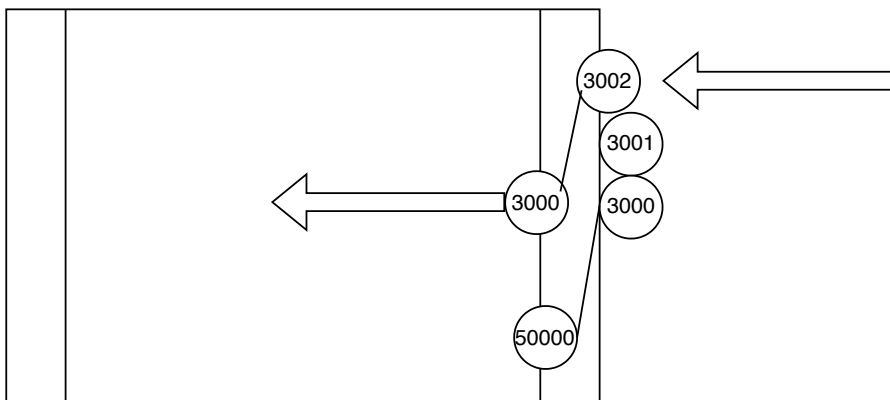
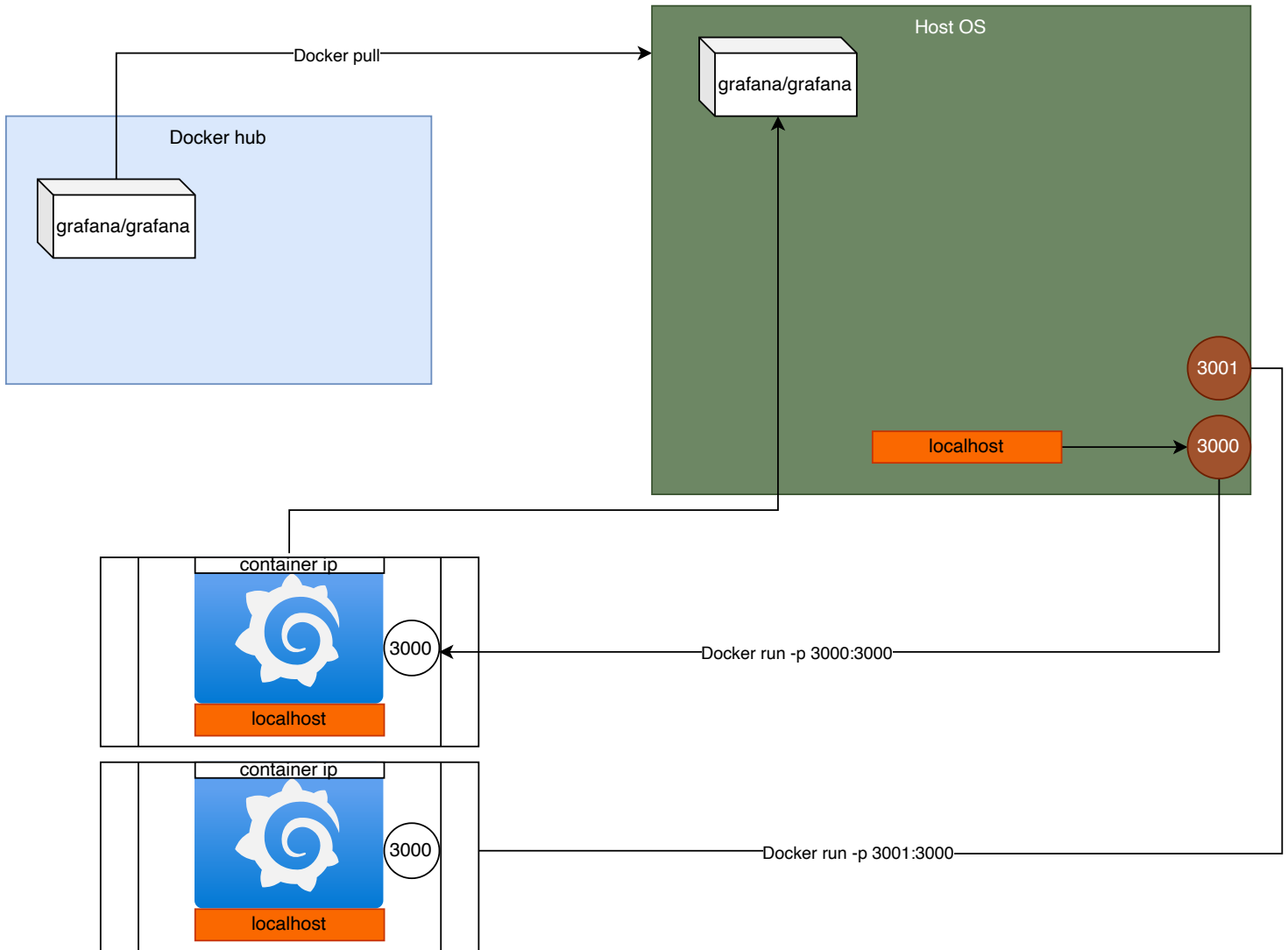
- one or more files missing in the prod
- software version mismatch
- different configuration setting

Docker Archithcture:

docker object  
containers networks  
volumes images



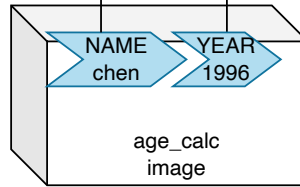
Running containers



environment variables in  
docker

```
FROM python:3.14.0a6-alpine3.21  
  
WORKDIR /applicaiton  
  
COPY code.py .  
  
ENV NAME=chen  
ENV YEAR=1996  
  
CMD [ "python3", "code.py" ]
```

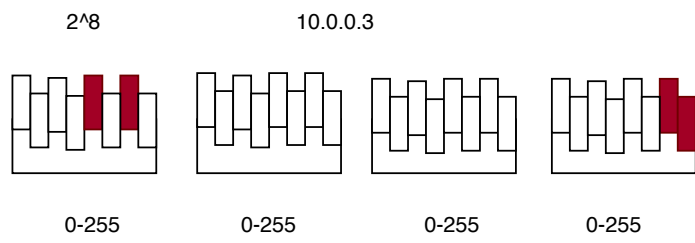
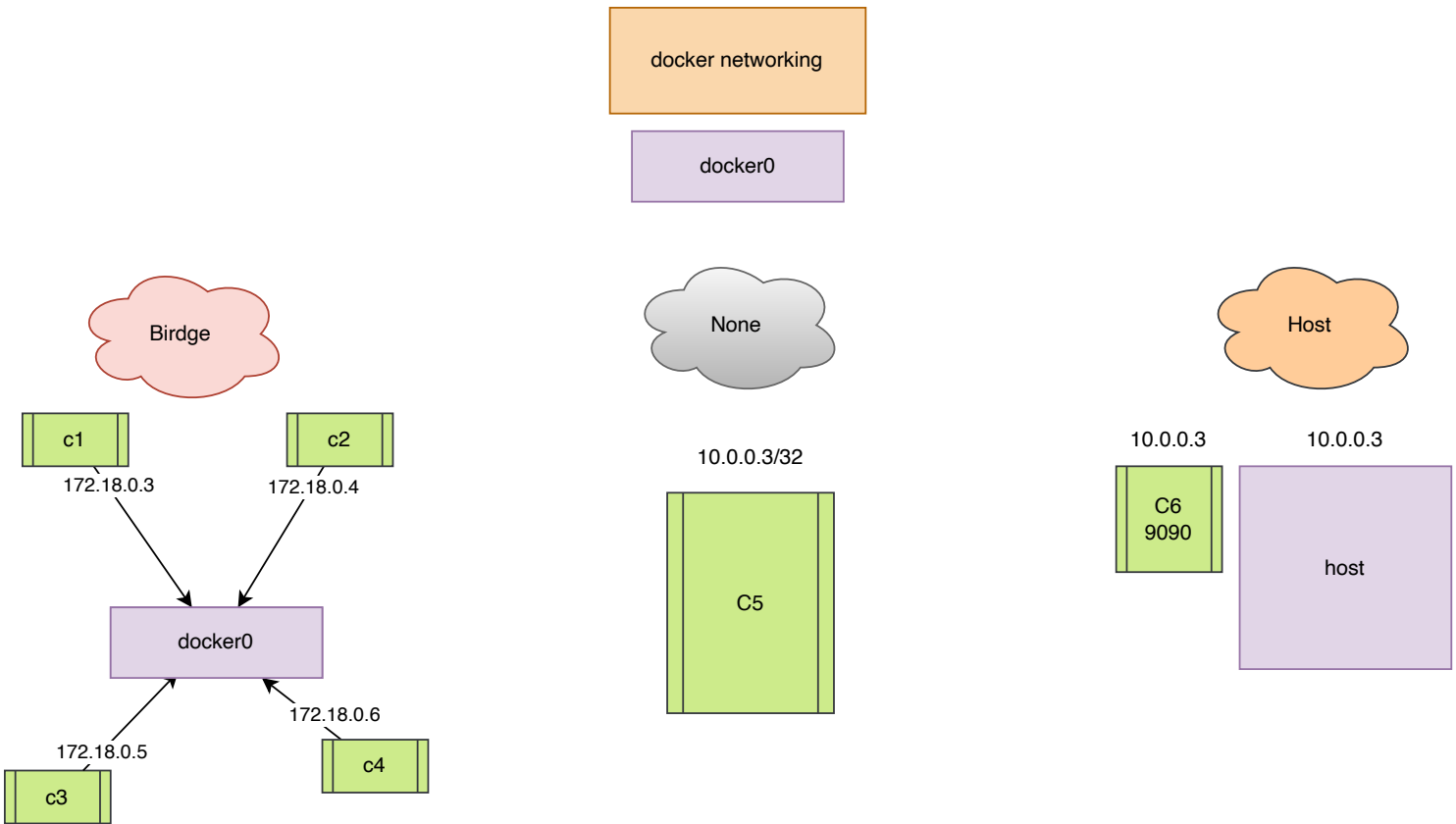
Dockerfile



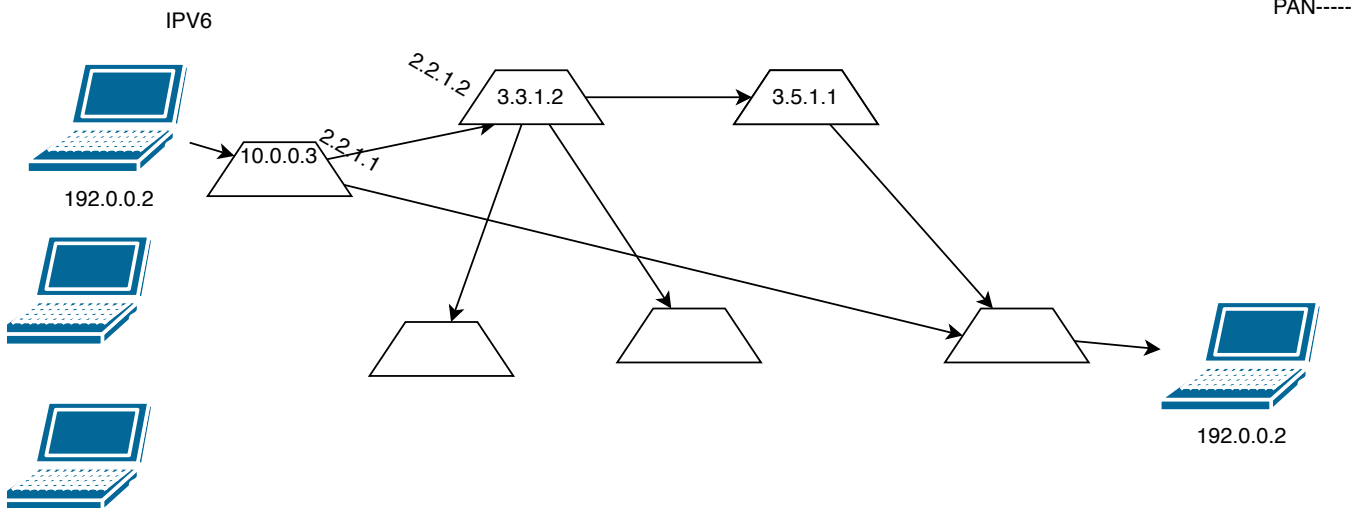
-e NAME=omer  
-e NAME=omer -e YEAR=1900

NAME=omer year=1996

NAME=chen year=1996



$256 \times 4 = 4,127,000,000$

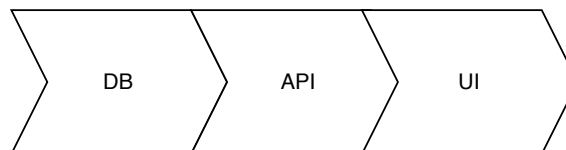
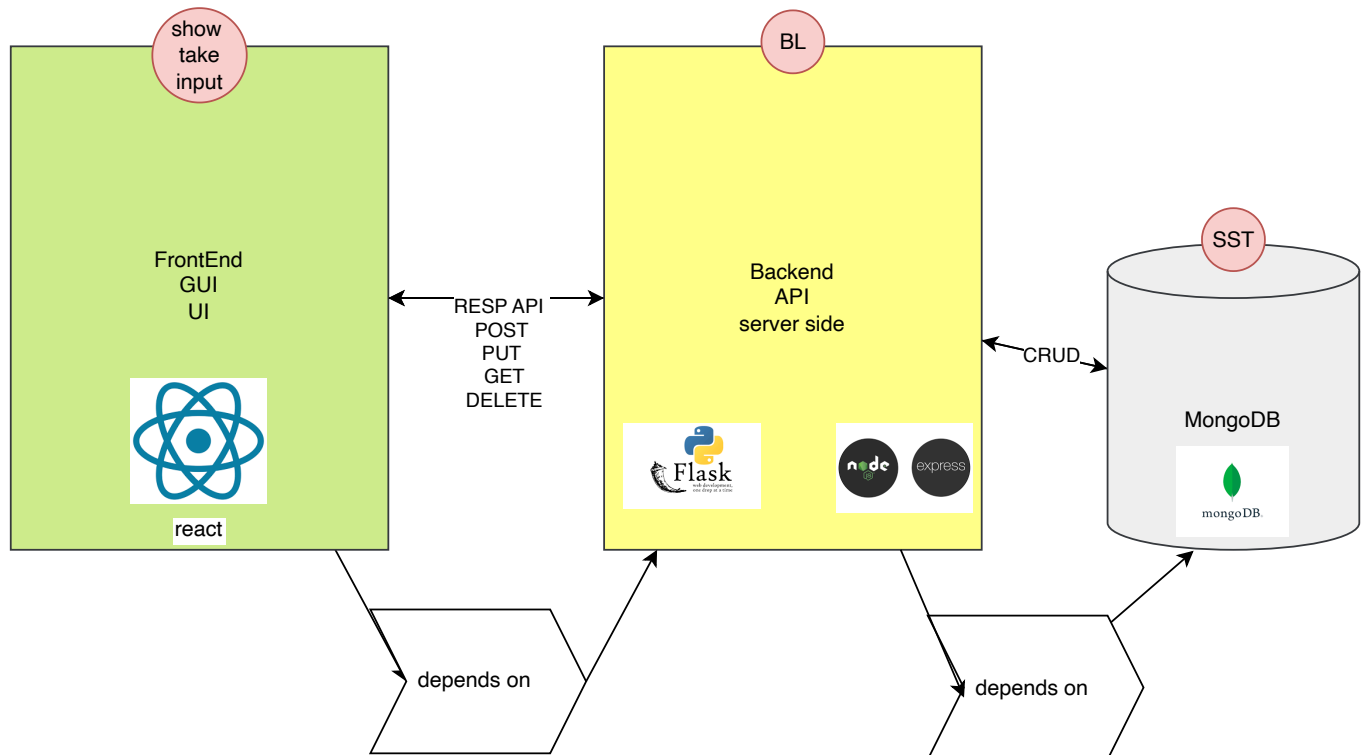


WAN  
LAN  
MAN-----  
PAN-----

## Docker full exercise

[https://github.com/hothaifa96/DevSecOps18/tree/docker/class\\_code/docker\\_networks](https://github.com/hothaifa96/DevSecOps18/tree/docker/class_code/docker_networks)

### TASK MANAGER APPLICATION



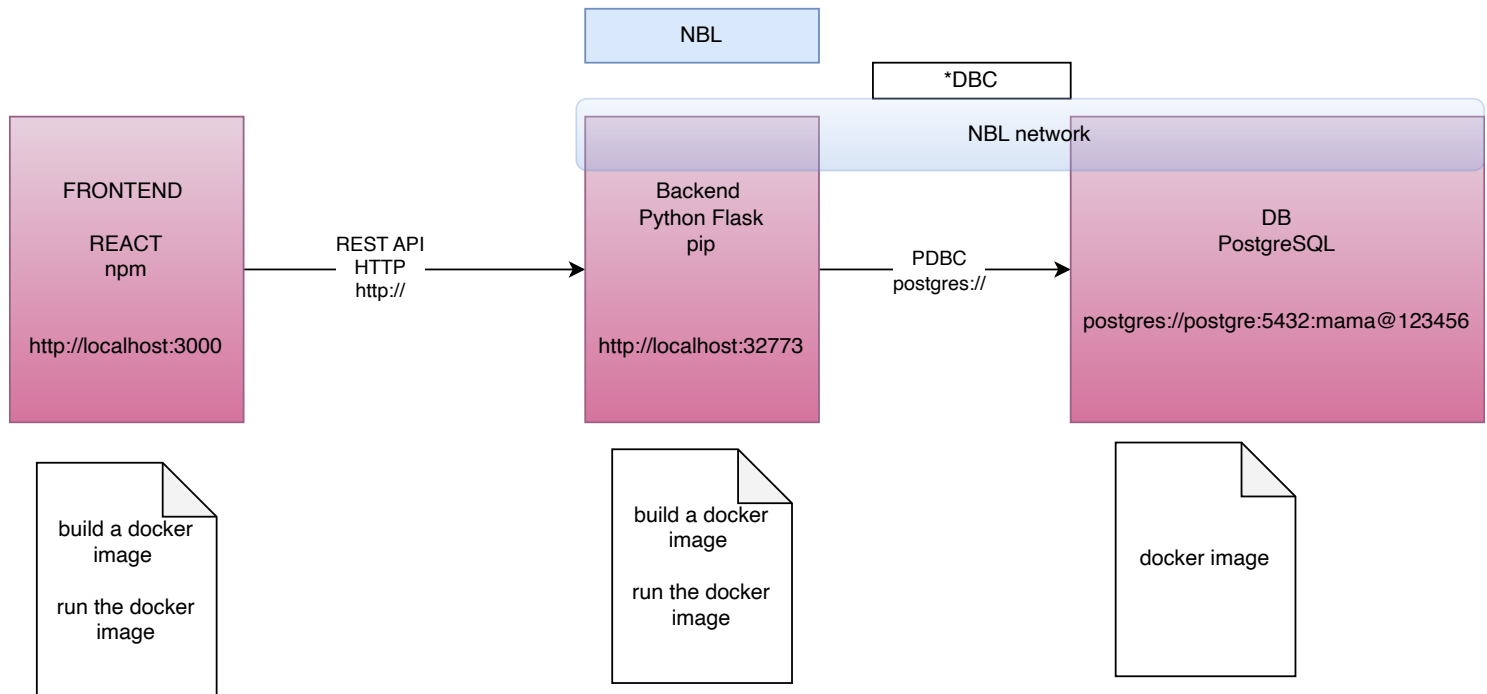
start a new mongodb container

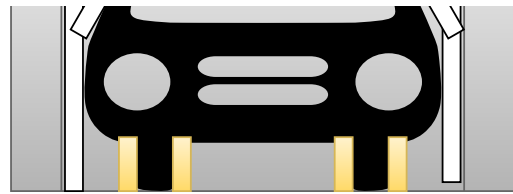
27017

publish port

network tast-m

network  
tast-m



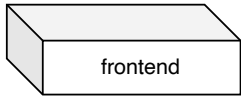


- OS + python env
- folder
- code files
- install 3rd party lib
- keep port open
- how to run the application

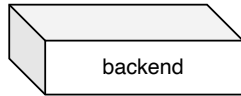
Dockerfile

|         |            |
|---------|------------|
| FROM    | USER       |
| WORKDIR | CMD        |
| WORKDIR | ENTRYPOINT |
| COPY    |            |
| ADD     |            |
| RUN     |            |
| ENV     |            |
| EXPOSE  |            |

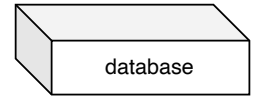




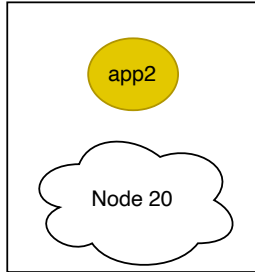
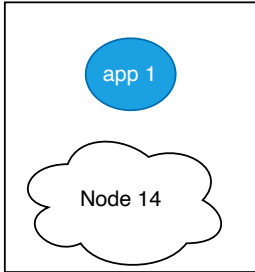
REACT  
Node 14



javascript  
Node 20



postgresql  
13



Virtual amachines



hypervisor



hypervisor



hypervisor

vmware,virtualbox

Contianer

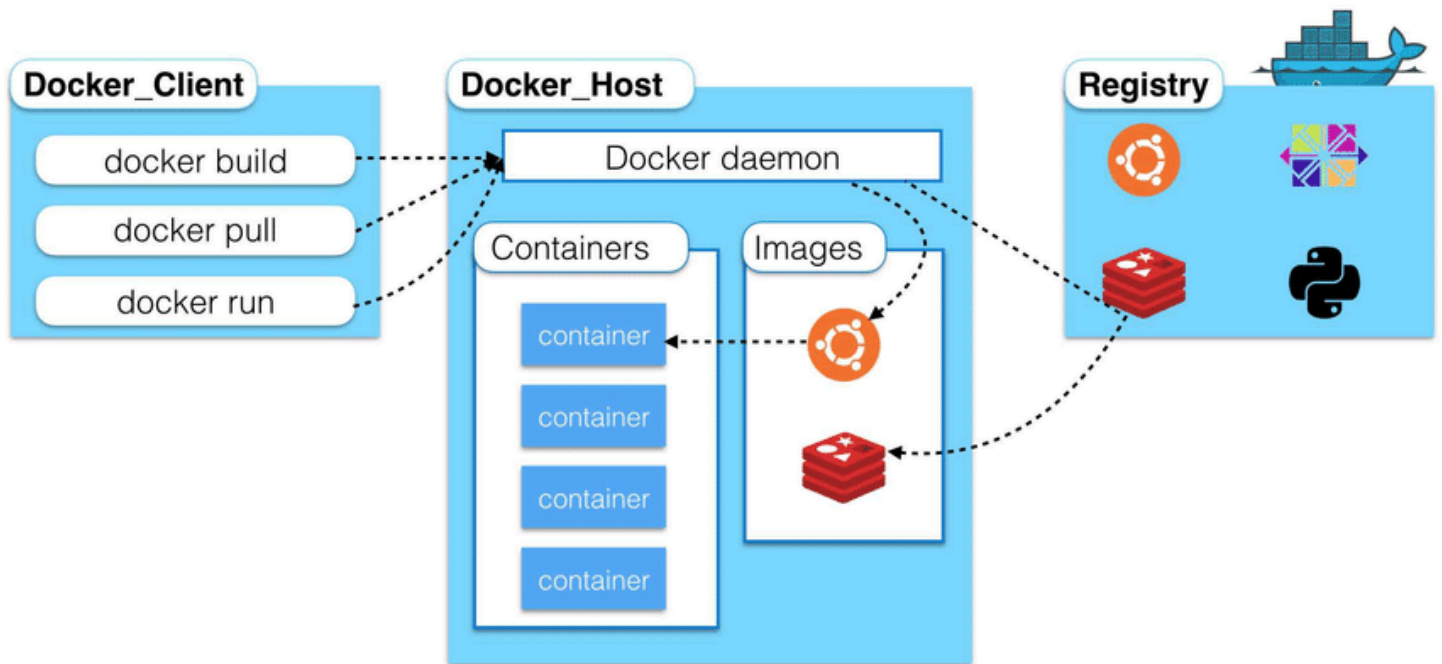
windows

Contianer

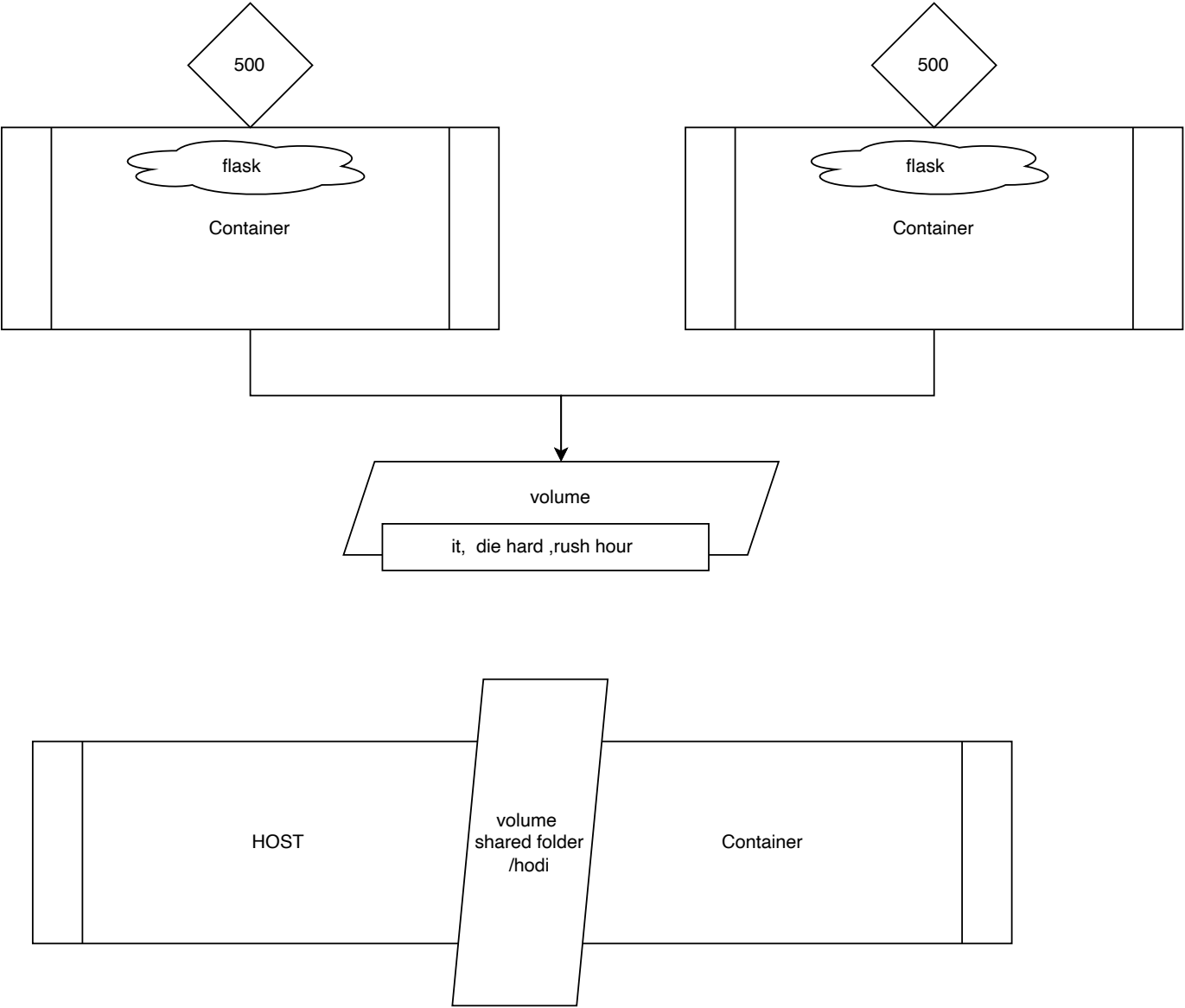
Linux

Contianer

MAC



volumes in docker

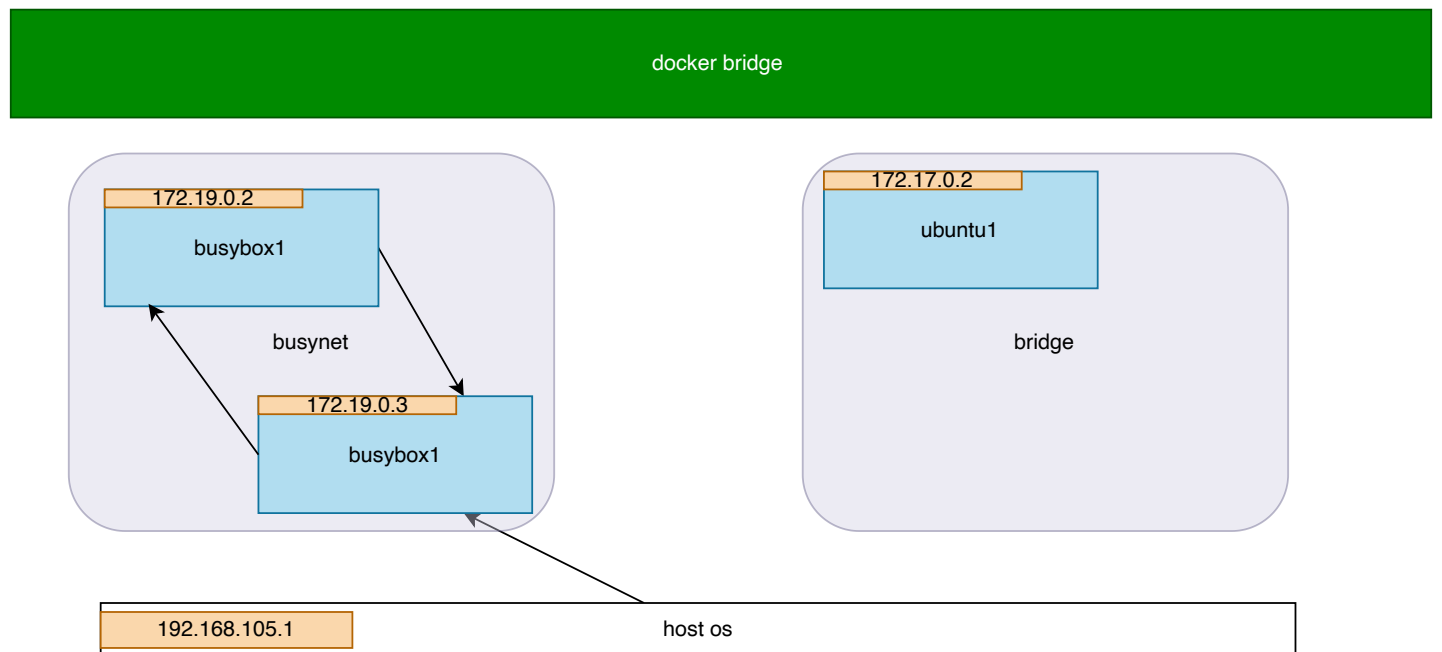
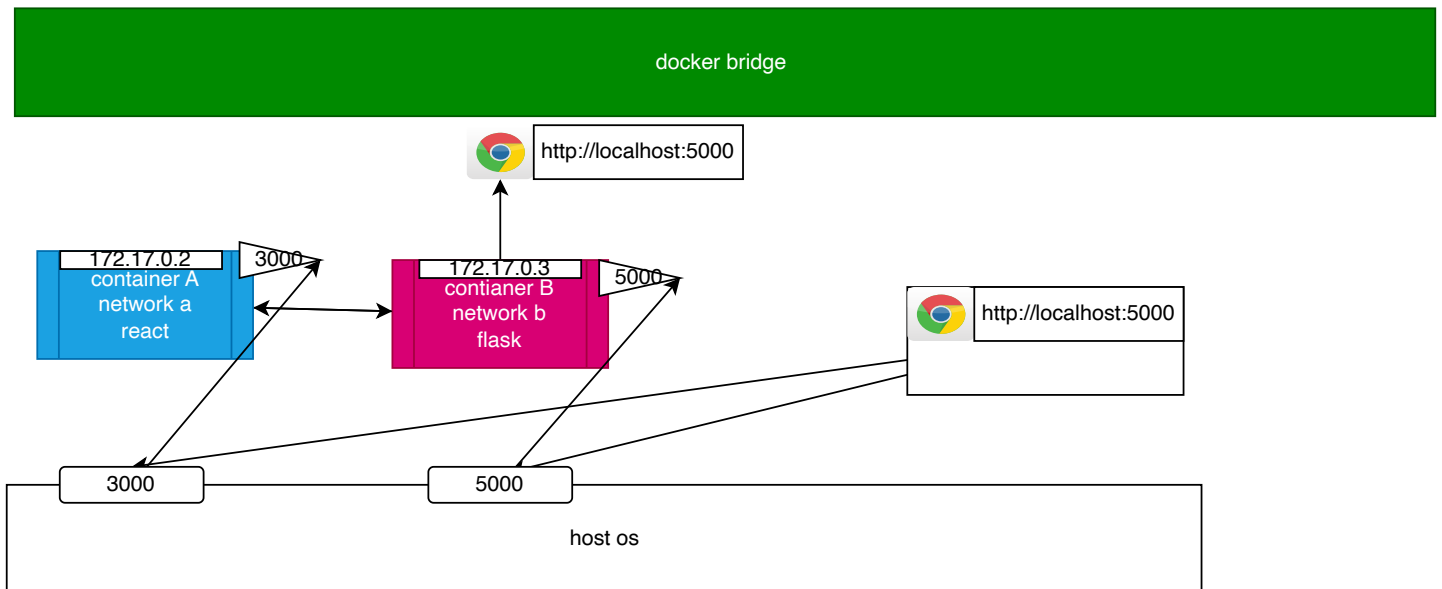


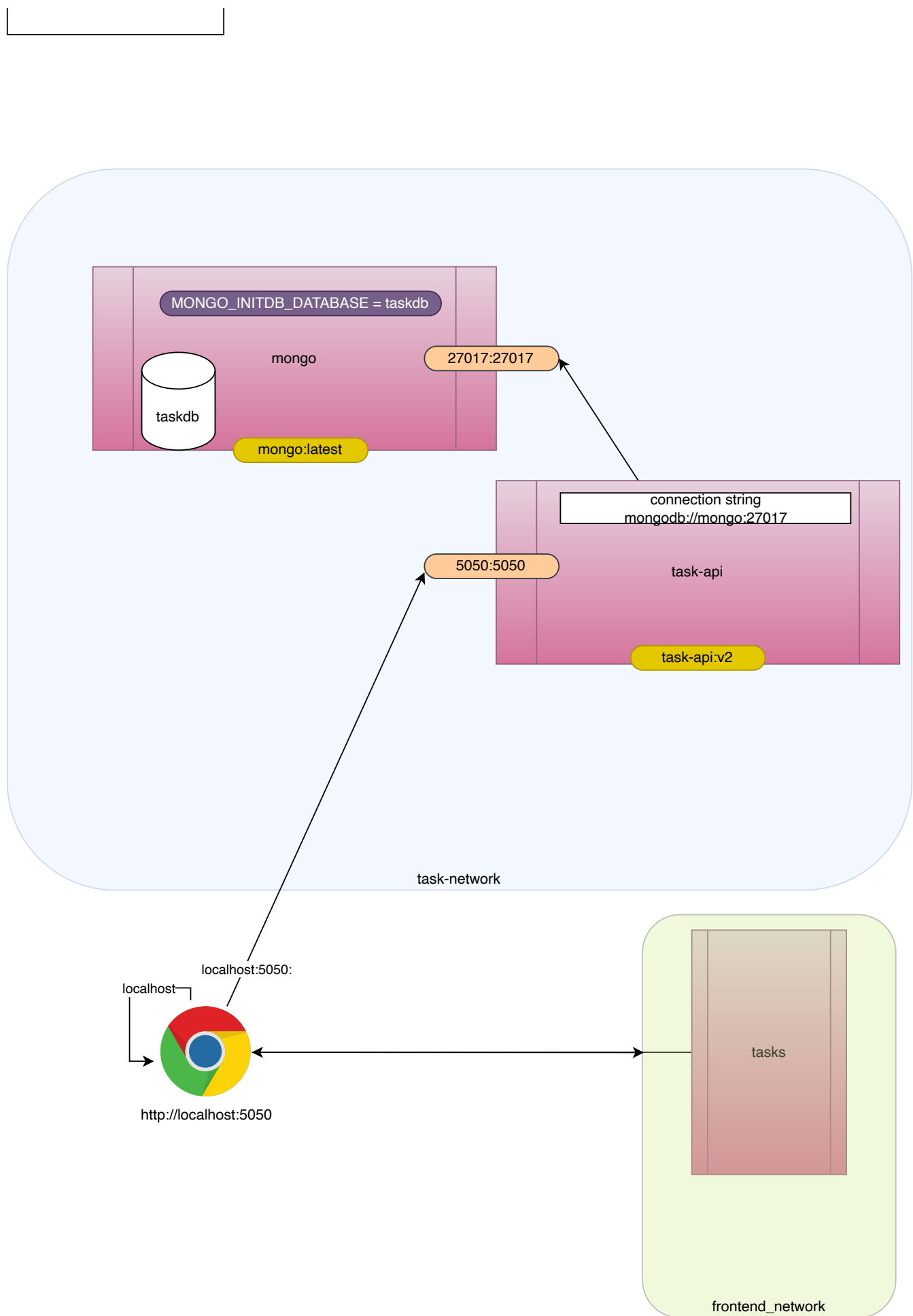
1.1.1.1/24

1.1.1.1/16

1.1.1.1/32

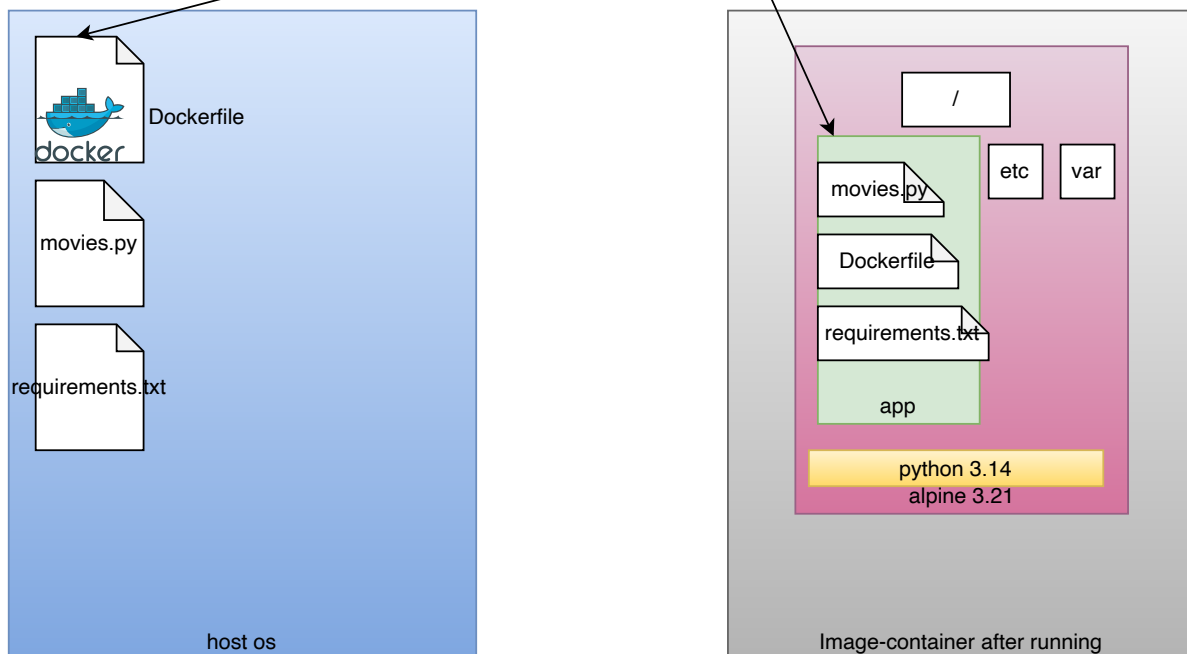
| network<br>name | machine<br>ip |
|-----------------|---------------|
|-----------------|---------------|



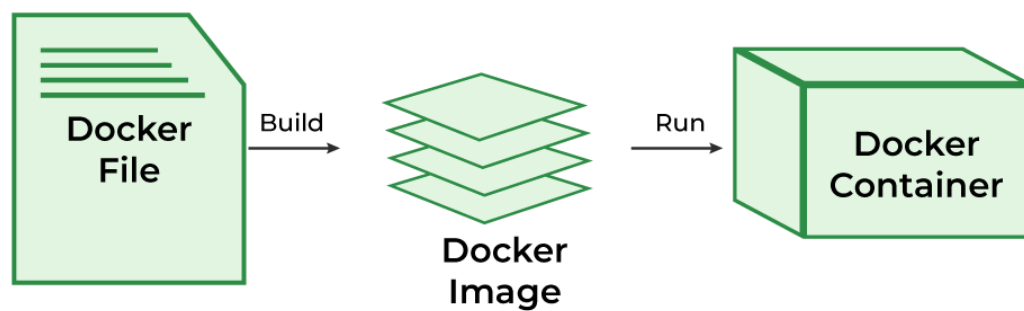
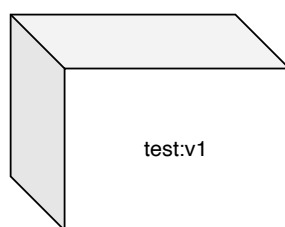




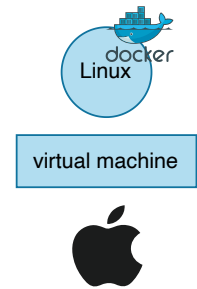
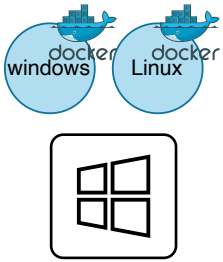
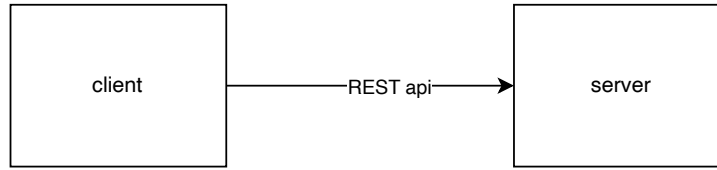
# COPY . .



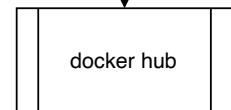
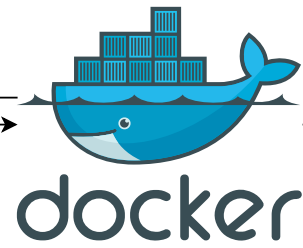
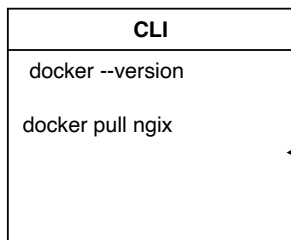
```
docker build -t test:v1
```



containers

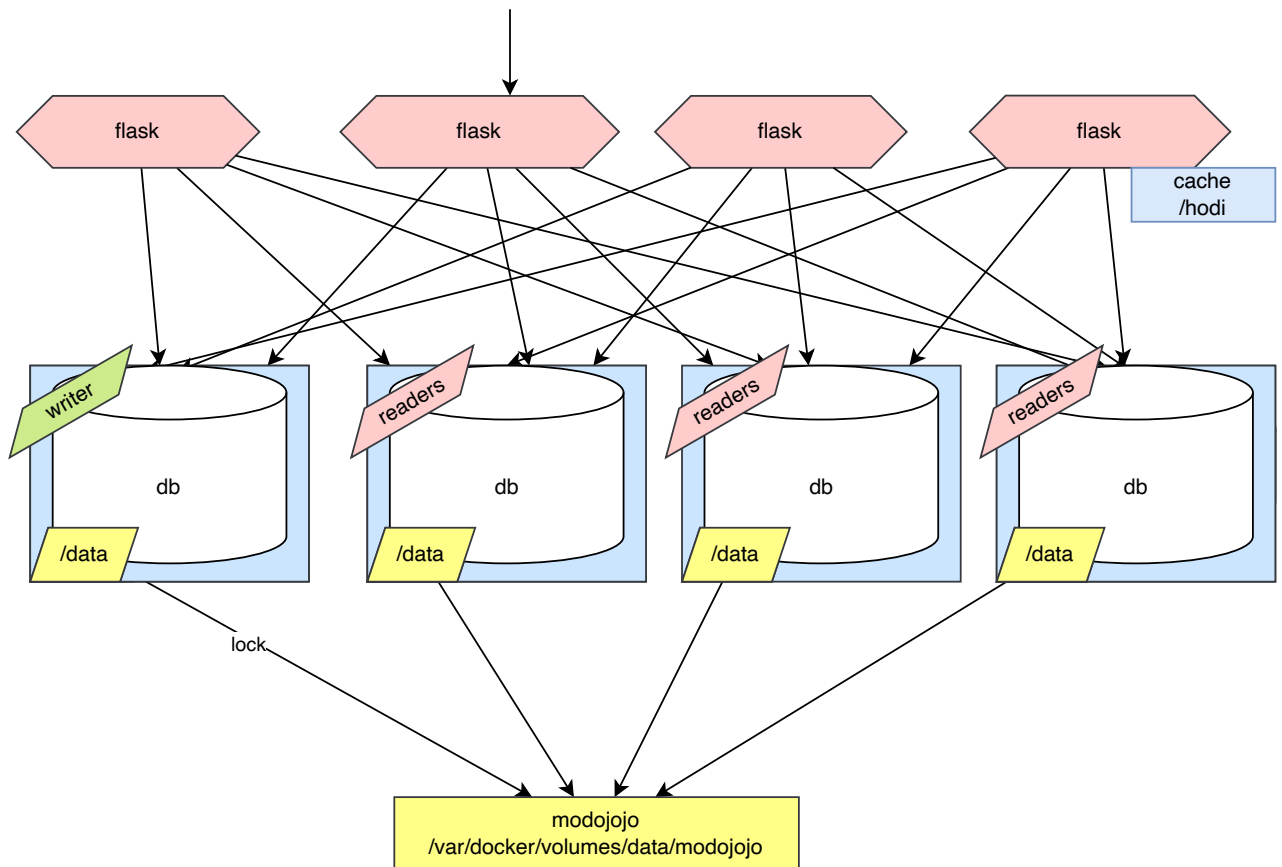
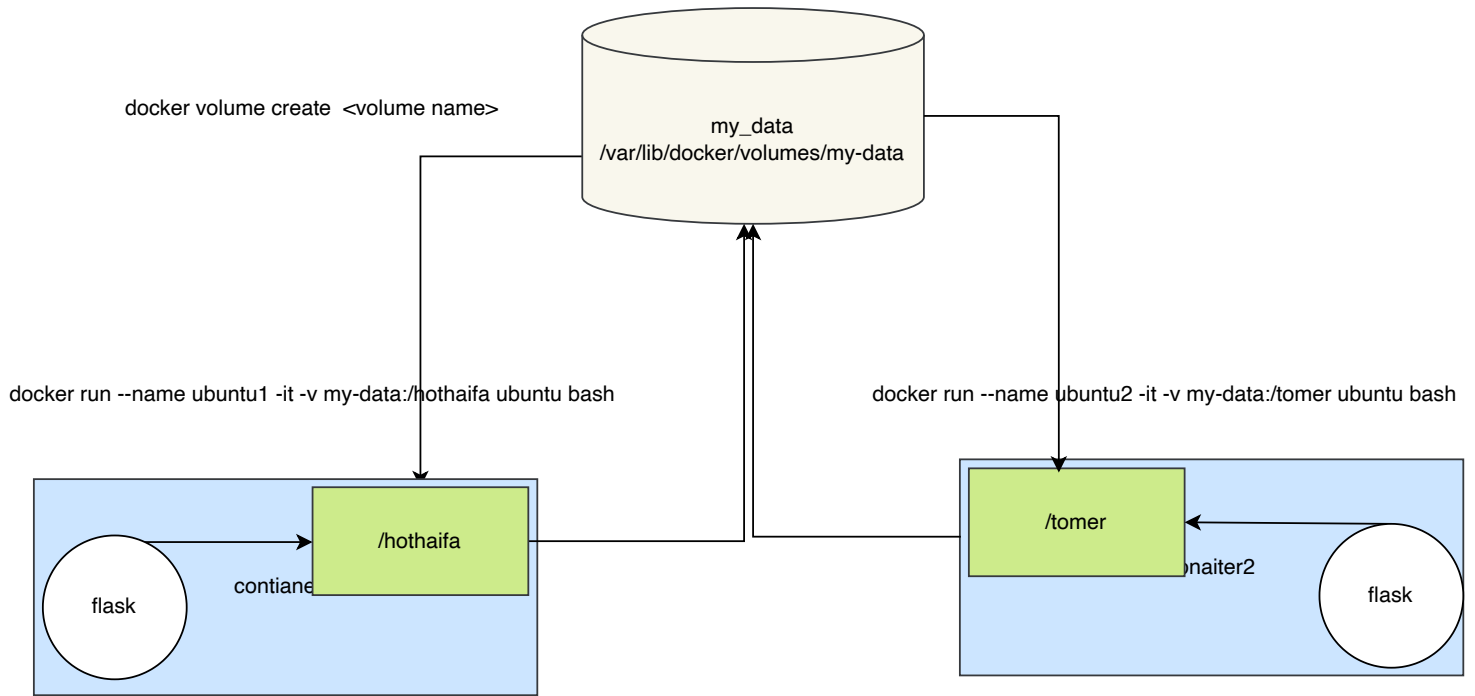


Docker demo

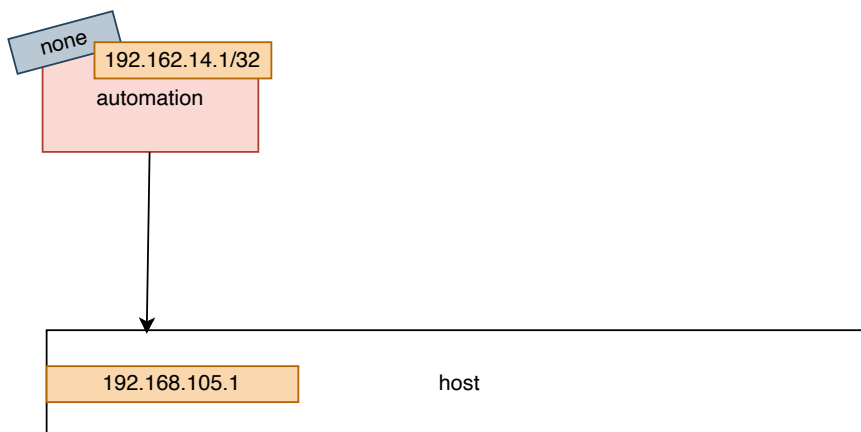
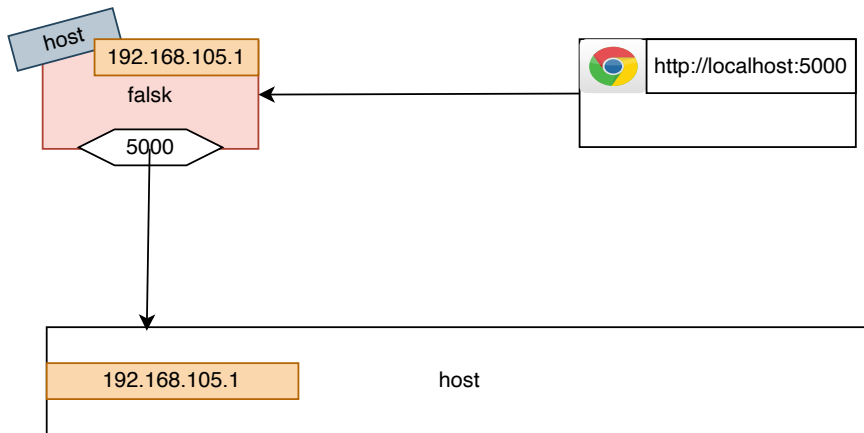


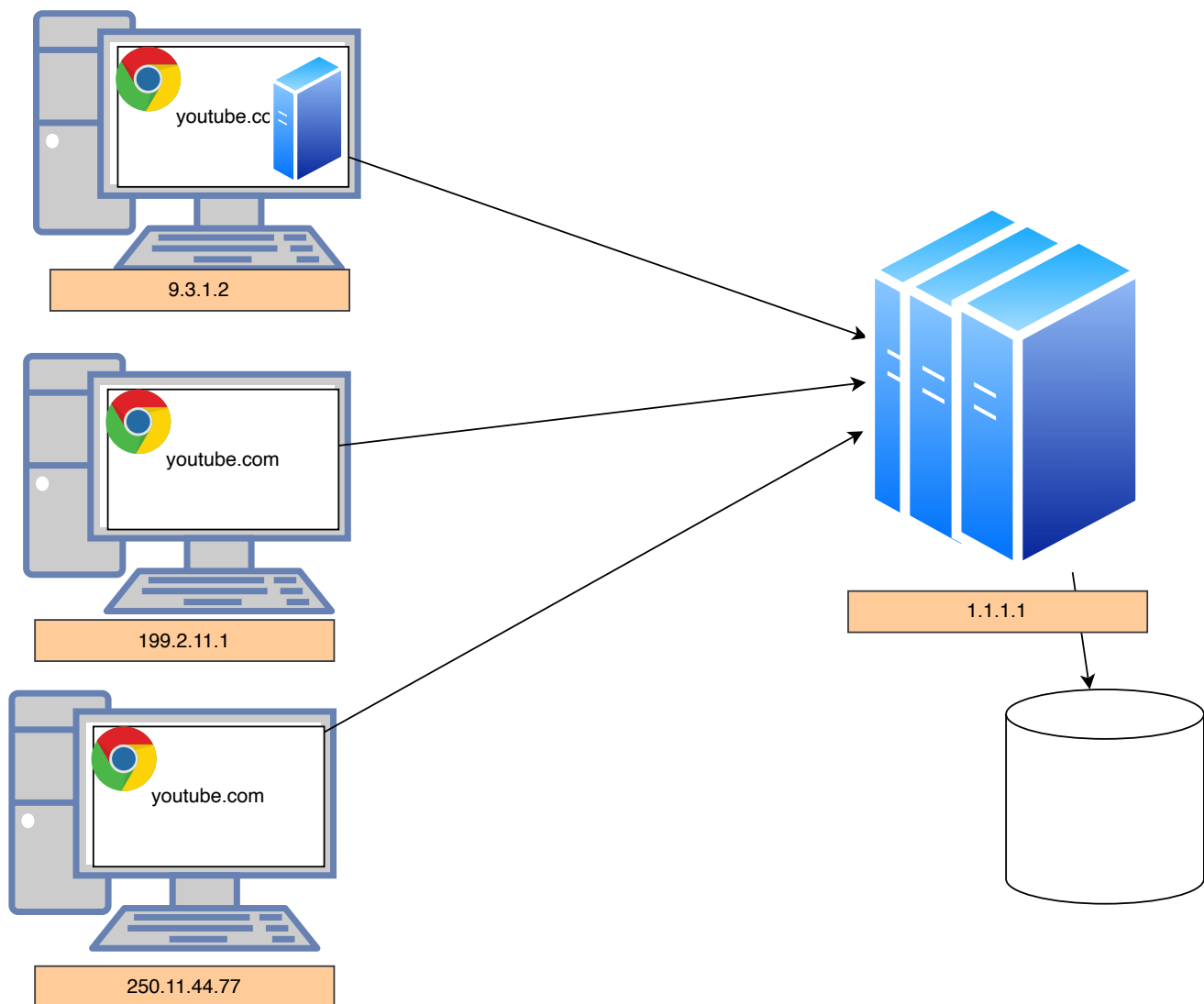






## docker Host network















# queue

